

Online Logger User Setup Guide

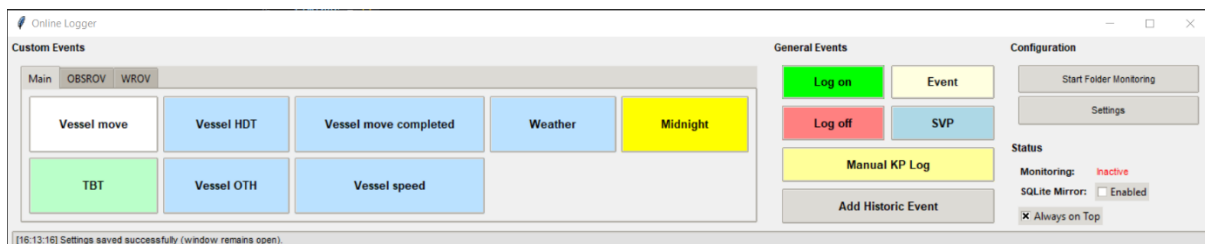
The Online Logger requires a few initial steps to define where it will read data from and where it will save the logged events.

Much of the setting changes can be done once setup through the GUI by right clicking. Things like adding buttons, editing buttons, adding tabs of buttons and such can all be done by right clicking at various points in the gui.

The program launches with a cmd window. This is a debugging window to help you trace errors – it should be left open, if you close it then you will also close the gui.

1 Initial Program Launch

Ensure you have a excel template setup with all the columns and headers you require. The headers must match the log program to work. If you accidentally change or delete a header then the program will not know where to place the data and you will get an error.



When you first run the program, the main window will appear, and the **Settings** dialog will open automatically. This dialog is where you'll configure all paths, column mappings, and custom buttons.

The Settings dialog uses a tab structure to organize the configuration:

Tab	Purpose
File Paths	Defines the main Excel Log file, data input folders, and the optional SQLite DB path.
Generated Fields	Configures the Excel column names for application-generated data (e.g., timestamps, UUIDs).
Static Fields	Defines data read from fixed, non-logging cells in the Excel file.

Tab	Purpose
Button Configuration	Customizes event buttons: text, log event description, source, and tab group.
Event Codes	Manages a list of standard codes that can be assigned to event buttons.
Monitored Folders	Configures non-TXT data folders (e.g., SVP, SSS) to log the latest file name.
Programmed Events	Configures automatic logging events (hourly KP, midnight).
Timezone	Sets the UTC offset for the "Local Time" column.
Database Sync	Configures automatic synchronization from Excel to the SQLite mirror.

2 Configure Essential File Paths (File Paths Tab)

The first step is to tell the program where your log and source data files are located.

2.1.1 Excel Log File

- Navigate to the **File Paths** tab.
- Click the "**Browse...**" button next to **Excel Log File**.
- Select your main log file (a blank or existing **.xlsx** or **.xlsb** file).

2.1.2 SQL DB

This is optional and does not need to be used if you don't want to write to an SQLite db from the program – Reach horizon spreadsheets manage the writing to SQL so this can be used as a backup if there is an issue with that. The SQL is a mirror of the excel workbook. It will take the headers from excel to make the table and then each time a event is logged it writes to excel and the SQL. Every X amount on minutes the DB is synced so any changes you made after the event button is pressed will be updated in the SQL – The UUID is the Primary key to know which row belongs to which in excel. If you are running this at the same time as using a horizon workbook then the horizon will complain since each time a sync takes place to the SQL, the workbook becomes locked for a second – horizon will pop up a cmd window with an error. You can reduce the errors by setting the automatic sync interval higher.

2.1.3 Navigation TXT Data Sources

You can configure up to five different data sources. The **Main TXT Data** source is typically your primary vessel navigation file.

1. In the **Main Vehicle Navigation (Main TXT Data)** section, click the "**Browse...**" button next to **Folder Path**.
2. Select the folder where your primary navigation .txt, .npd, or .csv files are saved.
3. (Optional) Enter a **Source Name** (e.g., "Vessel Nav") to give the source a user-friendly alias.
4. Repeat for any additional sources (TXT Source 2, 3, etc.) if needed.
5. The program uses the folder location you select to find the latest file and then the latest line in that file – For example in Qinsy you set the log file to create a new one each day, so you just point it at the parent directory and it will search in there for the latest file and pull the latest line.

2.1.4 Field Mapping for Data Sources

After setting the folder path(s), you must tell the program which piece of data in the file corresponds to which log column.

1. Click the "**Map Fields**" button for the source you want to configure ("**Main TXT**" to start). This opens the **Field Mapping** dialog.
2. Your data files are comma-separated. The table columns here correspond to the order of data in the file.
3. For each row:
 - **TXT Column:** Enter a user-friendly name for that piece of data (e.g., "KP", "Latitude").
 - **Excel Column:** Enter the exact header name in your Excel log where this data should be written (e.g., "KP", "Event", "Latitude"). This is case sensitive so make sure you match the header exactly.
 - **Skip?:** Check this box if the field exists in your data file but you **don't** want it written to Excel/DB.
4. (Optional) Click "**Preview Latest Data**" This will load the decoded fields from the selected text file for that object so you can check if you have the mapping set correctly.
5. Click "**Save Mappings**" to close the dialog and save the configuration.

6. You can define the mapping for each individual text file, or if they are the same you can press the '**Use Main Mapping**' button which will take the default mapping from the main text file. It's probably good to press this button anyway as it instantly adds a load of fields you can edit.
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3 Generated Fields

These are fields that the program generates itself – these need to have excel headers for each one to work – It is critical to have the header in excel for these to work.

4 Static Fields

You can add a number of static fields in this tab, these are fairly redundant but can be used if you have a static field you want to log such as project manager etc

5 Monitored Folders

This feature tracks the latest files created in various data folders to log the filename (or an 'X') in your log.

The **active logging threshold** is used as a way of looking to make sure that you are logging into the selected directory. If you set it to 15 seconds then it will only pull the file name if it's actively logging in the last 15 seconds. If you are testing that it works before actually logging data then you should set this value to a very high number or else the cell will fill with NA because it can't find a file in the set time frame. Basically it considers the logging as inactive.

It works as a QC to ensure you are logging but also stop the log being filled with a file name when you are not actually logging that file for the current task. SVP is exempt from this since it's not logging all the time while online

1. Navigate to the **Monitored Folders** tab.
2. For each folder listed (e.g., Qinsy DB, SVP):
 - Click the "... " button next to **Monitor Path** and select the folder.
 - Set the **Excel Column** (the column header in your log file) where the filename will be written.

- Set the **File Ext.** (e.g., **npd**). Leave blank to track any file in the folder. This ensures that the log only pulls the file type you are interested in, such as.npd
 - **Log 'X'?:** Check this if you want an "X" logged instead of the full filename when a file is active.
 - **Log Ext:** This logs the file extension – for the SBD files the name can make excel think the format should be in scientific format so it doesn't display correctly. Having the .sbd at the end negates this. But you can do this to any monitored folder to make sure that its getting the correct file
 - **Skip?:** Check this if you do not want to monitor this specific folder.
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6 Customize Event Buttons (Button Configuration Tab)

You can define up to 50 custom event buttons for quick logging. Each button can be edited by right clicking it in the GUI, but you can get a overview here of all buttons to make changes

1. Set the **Number of Custom Buttons** you need and click "**Update List.**"
 2. For each custom button:
 - **Button Text:** What appears on the button (e.g., "Log On").
 - **Event Text (for Log):** The detailed description written to the "Event" column (e.g., "Log On").
 - **Event Source:** Select the TXT Data Source (e.g., "Main TXT") whose navigation data should be logged when the button is pressed. Select "**None**" if no navigation data is needed.
 - **Event code** – Shorted event code assigned in the event code tab – this can be used in the future for more automated reporting
 - **Tab Group:** Enter a name to group buttons onto a separate tab on the main window (e.g., "WROV").
 - **Set the colour of the button and the text** – this will be reflected in the row in excel
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7 Event Code

In here you can define several event codes that you can assign to each button – Something along the lines of L-ON for Log on – this directory creates its own settings file of event codes so you can use a old setting file for this rather than having to input new codes each time.

Programmed events

8 Starting Monitoring and Logging

After saving your settings, you are ready to begin work

8.1 Start Monitoring

1. On the main window, click the "**Start Folder Monitoring**" button (in the Configuration section on the right). This function scans all the folders you have selected to monitor and then holds the latest 5 files it finds in a cache – this speeds up the process of logging data because it only needs to look at 5 files rather than all the files in the folder. The initial start monitoring may take a while if you have lots of folder to look into selected.
2. The **Monitoring** indicator should change to **Active** (green), and the status bar at the bottom should update. This initiates the background process that scans the monitored folders.
3. If you change a directory or want to monitor a new directory then you need to stop monitoring, add/change a file path you want, then start monitoring again.

9 Programmed events

The programmed events are:

- midnight position (automatically adds a midnight position at midnight)
- Hourly KP log – This needs further testing but it should consider the line name and accumulate the KP distance covered between the last KP event which is 1 hour, unless you manually press the manual KP log on the main gui. This only works if you have the KP column populated with data – if not then it will not return anything
- Log off distance/speed calculation – this looks at the time and KP of the last log on when you log off and automatically fills the cell with the distance covered and the average speed. If there isn't a KP associated with log on and log off then the function won't be called and it will just say Log off

10 Time zone

This is the time difference between UTC and Local time, then it fills in the Local Date/Time Column for each entry with the local time

11 Database Sync

The database Sync is only valid if you are using the program to log to an SQL. In here you set the automatic sync interval which will mirror any changes you make to the excel log to the SQL. Since the SQL is logged instantly on a button press, this doesn't always mirror what you may change the event text to for example. This function ensures that changes are pushed though at a defined interval and the field log viewers are kept up to date

12 Project Settings

Use Projects to save and reuse all configuration for a job or vessel. A project file (.json) holds your monitored folders, mappings, button layouts/colors, programmatic events, and event codes in one place.

Where to find it:

- Open Settings → Projects.
- The Current Project section shows the active project path (read-only).

Controls

- **Browse...** : Select an existing project (.json). The path field updates; nothing is applied until you Load.
- **Load Project:** Loads the selected file and makes it the active project. The app remembers the last loaded project for next start.
- **Save Project:** Saves all current settings to the active project path. If no project is active, you'll be prompted to use Save As.
- **Save As...**: Creates a new project file (.json) and makes it the active one. Use this to branch settings per client/job.
- **Restore Blank Project (red):** Loads a clean template into memory (no files are overwritten). The active project path is cleared; use Save As to persist.

How it behaves

- One project file contains everything, including event codes (no separate codes file needed).
- The last loaded project is remembered automatically and reloaded on the next launch.

- After restoring the blank project, the structure viewer shows default values; use Save As to store it.
- Changes made anywhere in Settings only persist to disk when you Save (or Save As) the project.

Tips

- Keep a separate project per scope (e.g., vessel, field, or client/job) to avoid accidental cross-use.
- Store projects under the app's settings folder for easy backup, or version them in source control if desired.

13 Logging Events

13.1 General Events:

- **Log on / Log off:** Use these to record the start and end of a logging session. "Log off" will automatically calculate distance traveled and speed if enabled in settings.
- **Event:** Logs an event using the navigation data from the selected source (default: Main TXT).
- **SVP:** Logs a standard event and inserts the filename of the latest SVP file found in the monitored SVP folder.

13.2 Custom Events:

Click any of your configured custom buttons (e.g., "Boulder") to log a custom event using the source and text defined in the settings.

13.3 Historic Event:

Use this button (in the General Events section) to retroactively log an event from a specific date/time by manually searching a past log file for a time string.

13.4 Editing Buttons on the Fly

You can quickly change a button's text, event description, source, or color without opening the main Settings window.

1. **Right-click** on any **General Event** or **Custom Event** button.
2. Select **"Edit [Button Name] Settings..."**.

3. Make your changes in the small dialog, then click "**Save**". The button will instantly update.
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14 Saving and Closing

When you are done, close the application by clicking the standard window close button (X). The application will automatically stop all monitoring processes.